

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (2 Question * 100 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
Student Name:	N. NDKShagna	Reg. No.:	192472205
Section / Batch:	CSE-A1 / 2024	Date:	15/07/26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas, Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

☉ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

☉ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems

Topic 3: Architecture of Data Warehouse

- ☉ Focus Areas:
 - Data warehouse definition
 - Three-tier architecture
 - ETL process
 - Data marts & metadata
 - Benefits in decision-making

Topic 4: OLAP and Data Cubes

- ☉ Focus Areas:
 - OLAP concept
 - Data cube structure
 - OLAP operations (slice, dice, roll-up, drill-down)
 - Applications in analytics

Topic 5: Dimensional Data Modeling

- ☉ Focus Areas:
 - Fact and dimension tables
 - Star schema vs snowflake schema
 - Schema design process
 - Real-world example

Topic 6: Query Processing & BI Tools

- ☉ Focus Areas:
 - Query processing stages
 - Query optimization
 - BI tools overview
 - Reporting and visualization
 - Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

- ☉ Focus Areas:
 - Introduction to KNIME
 - Workflow components (nodes, connections)
 - Performing OLAP operations
 - Sample application/demo
 - Report generation

Code of Conduct / 192432205
 I, N. MOKESH (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
 Signature of the Student: N. MOKESH

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____

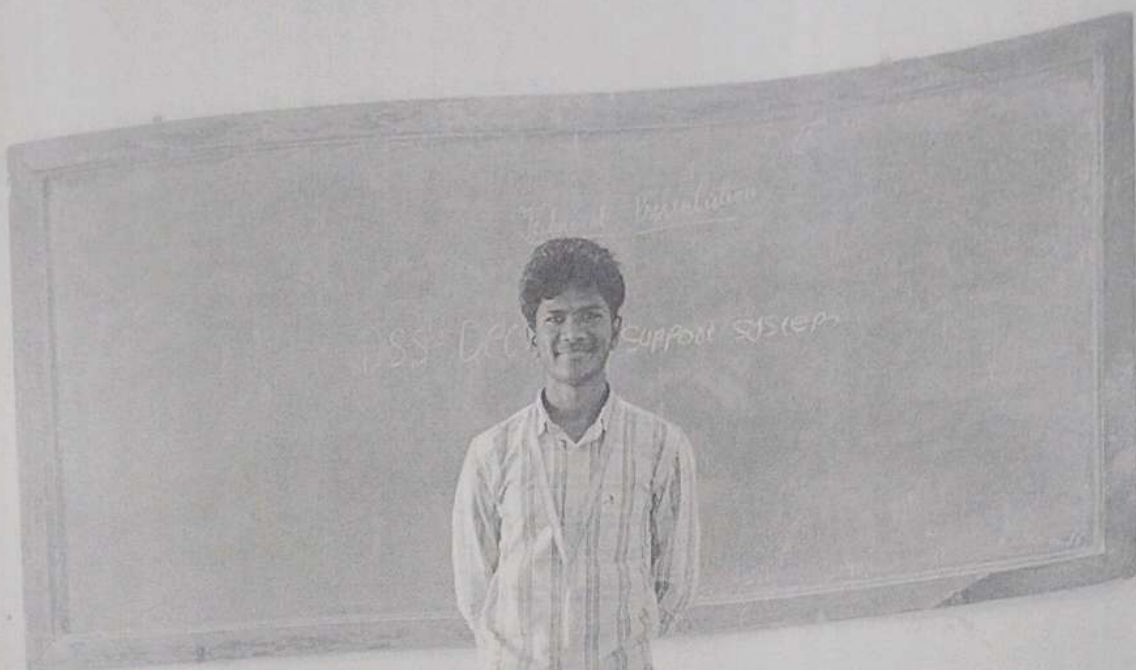
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GPS Map Camera

Kuthambakkam, Tamil Nadu, India

Cse & It Department, Kuthambakkam, Tamil Nadu 602105, India

Lat 13.025829° Long 80.016635°

Wednesday, 15/07/2026 11:53 AM GMT +05:30

Google

Data Warehouse Architecture

How organizations store and use data to make better decisions

DB

What is a Data Warehouse?

A data warehouse is a large central store that collects data from many different places in a company. The data is cleaned, organized, and kept so people can study it and find useful patterns over time.

- **Subject-oriented:** Organized by topic, like sales or customers
- **Integrated:** Combines data from many source systems
- **Time-variant:** Keeps history to show change over time
- **Non-volatile:** Data stays put once loaded; it is not deleted



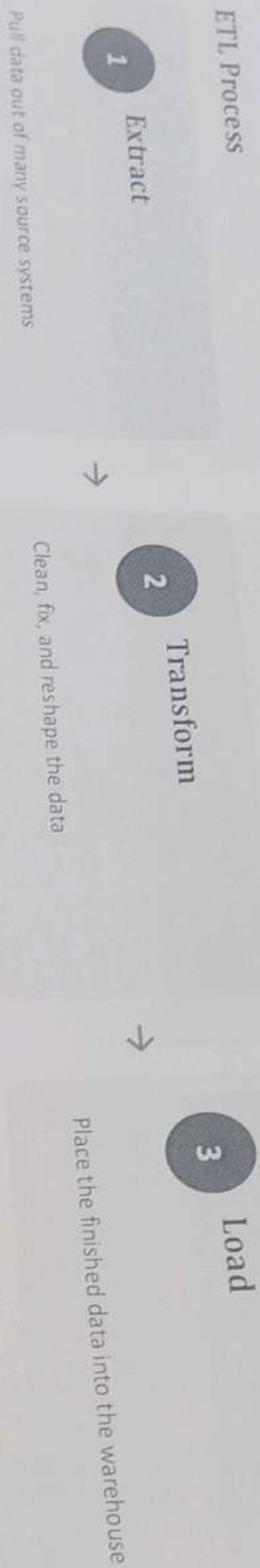
Benefits for Decision-Making

- ✓ One clear source of truth for the whole company
- ✓ Faster reports and dashboards for managers
- ✓ Easy to spot trends and compare past performance
- ✓ Better, faster, data-backed business decisions
- ✓ Frees up daily systems from heavy report work

ETL Process, Data Marts & Metadata

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ETL Process



M Data Marts

A data mart is a smaller, focused slice of the warehouse built for one team, such as Sales or Finance. It holds only the data that team needs, so their reports run faster and stay simple.

i Metadata

Metadata is "data about the data." It explains where each piece of data came from, what it means, and how it was changed. This acts like a map, helping users find and trust the right information.

Three-Tier Architecture

Data moves upward through three layers, from raw sources to the reports people see

TIER 1 Bottom Tier

Data Sources & ETL

Raw data lives here: sales systems, databases, files. The ETL process cleans and loads it into the warehouse.



TIER 2 Middle Tier

OLAP Server

An OLAP engine organizes the warehouse data into cubes, making it fast to slice, sum, and compare.



TIER 3 Top Tier

Front-End Tools

Users see the results here: dashboards, reports, and query tools used for analysis and decisions.

Data flows from Tier 1 → Tier 2 → Tier 3, becoming easier to understand at each step.

✓
RK